

CSE 3204
Formal Language, Automata and Computability



Lecture 11
Context Free Grammars, Parse Tree

Constructing a Parse Trees

- Let CFG $G = (V, T, P, S)$. The parse trees for G have following properties:
 - Each interior node is labeled by a variable in V .
 - Each leaf is labeled by either a variable a terminal or ϵ . If the leaf is labeled ϵ then it must be the only child of its parent.
 - If an interior node is labeled A and its children are labeled $X_1, X_2, \dots, X_k$ from the left then

$$A \rightarrow X_1 X_2 \cdots X_k$$

Note that the only time one of the X 's can be ϵ is if that is the label of the only child, and $A \rightarrow \epsilon$ is a production of G .

Examples of Parse Trees

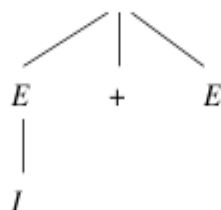


Figure 5.4: A parse tree showing the derivation of $I + E$ from E

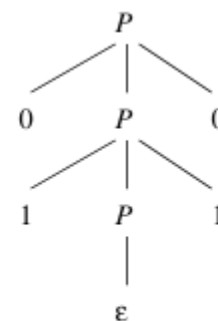


Figure 5.5: A parse tree showing the derivation $P \xRightarrow{*} 0110$

1. $E \rightarrow I$
2. $E \rightarrow E + E$
3. $E \rightarrow E * E$
4. $E \rightarrow (E)$
5. $I \rightarrow a$
6. $I \rightarrow b$
7. $I \rightarrow Ia$
8. $I \rightarrow Ib$
9. $I \rightarrow I0$
10. $I \rightarrow I1$

1. $P \rightarrow \epsilon$
2. $P \rightarrow 0$
3. $P \rightarrow 1$
4. $P \rightarrow 0P0$
5. $P \rightarrow 1P1$

Figure 5.1: A context-free grammar for palindromes

Yield of a Parse Tree

- If by concatenating the leaves of any parse tree from the left we get a string called the yield of the tree. It is a string that is derived from the root variable.
 1. The yield is a terminal string. That is, all leaves are labeled either with a terminal or with ϵ .
 2. The root is labeled by the start symbol.

Yield of a Parse Tree

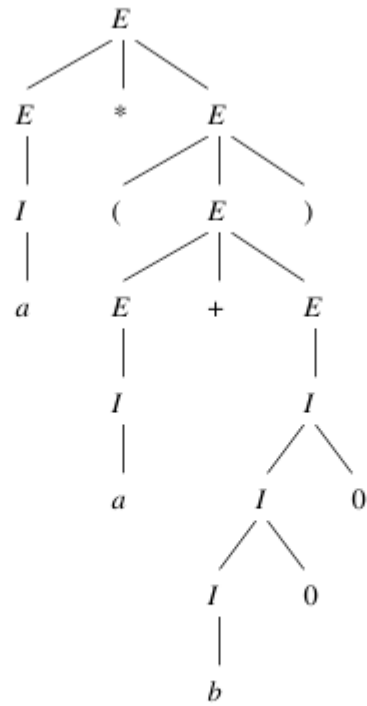


Figure 5.6: Parse tree showing $a * (a + b00)$ is in the language of our expression grammar

Review of Tree Terminology

We assume you have been introduced to the idea of a tree and are familiar with the commonly used definitions for trees. However, the following will serve as a review.

- Trees are collections of *nodes*, with a *parent-child* relationship. A node has at most one parent, drawn above the node, and zero or more children, drawn below. Lines connect parents to their children. Figures 5.4, 5.5, and 5.6 are examples of trees.
- There is one node, the *root*, that has no parent; this node appears at the top of the tree. Nodes with no children are called *leaves*. Nodes that are not leaves are *interior nodes*.
- A child of a child of a $\dots$ node is a *descendant* of that node. A parent of a parent of a $\dots$ is an *ancestor*. Trivially, nodes are ancestors and descendants of themselves.
- The children of a node are ordered “from the left,” and drawn so. If node N is to the left of node M , then all the descendants of N are considered to be to the left of all the descendants of M .

Inference, Derivation and Parse Tree

- Given a grammar $G = (V, T, P, S)$, the following are equivalent:

1. The recursive inference procedure determines that terminal string w is in the language of variable A .

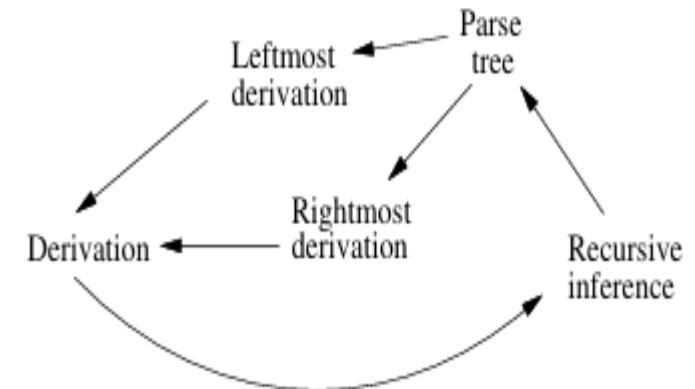
2. $A \xRightarrow{*} w$.

3. $A \xRightarrow[tm]{*} w$.

4. $A \xRightarrow[rm]{*} w$.

5. There is a parse tree with root A and yield w .

2, 3, 4 and 5 are also equivalent for string w with some variables in it.



From Inference to Parse Tree

Theorem 5.12: Let $G = (V, T, P, S)$ be a CFG. If the recursive inference procedure tells us that terminal string w is in the language of variable A , then there is a parse tree with root A and yield w .

PROOF: The proof is an induction on the number of steps used to infer that w is in the language of A .

BASIS: One step. Then only the basis of the inference procedure must have been used. Thus, there must be a production $A \rightarrow w$. The tree of Fig. 5.8, where there is one leaf for each position of w , meets the conditions to be a parse tree for grammar G , and it evidently has yield w and root A . In the special case that $w = \epsilon$, the tree has a single leaf labeled ϵ and is a legal parse tree with root A and yield w .

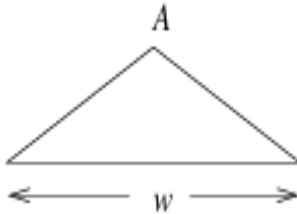


Figure 5.8: Tree constructed in the basis case of Theorem 5.12

INDUCTION: Suppose that the fact w is in the language of A is inferred after $n + 1$ inference steps, and that the statement of the theorem holds for all strings x and variables B such that the membership of x in the language of B was inferred using n or fewer inference steps. Consider the last step of the inference that w is in the language of A . This inference uses some production for A , say $A \rightarrow X_1 X_2 \cdots X_k$, where each X_i is either a variable or a terminal.

We can break w up as $w_1w_2\cdots w_k$, where:

1. If X_i is a terminal, then $w_i = X_i$; i.e., w_i consists of only this one terminal from the production.

From Inference to Parse Tree

2. If X_i is a variable, then w_i is a string that was previously inferred to be in the language of X_i . That is, this inference about w_i took at most n of the $n+1$ steps of the inference that w is in the language of A . It cannot take all $n+1$ steps, because the final step, using production $A \rightarrow X_1X_2 \cdots X_k$, is surely not part of the inference about w_i . Consequently, we may apply the inductive hypothesis to w_i and X_i , and conclude that there is a parse tree with yield w_i and root X_i .

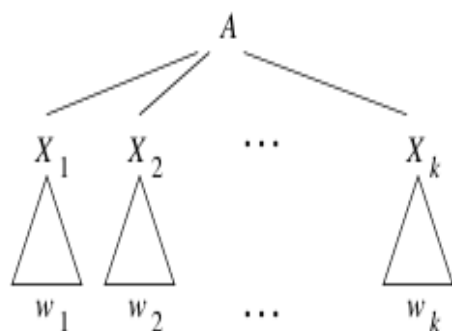


Figure 5.9: Tree used in the inductive part of the proof of Theorem 5.12

We then construct a tree with root A and yield w , as suggested in Fig. 5.9. There is a root labeled A , whose children are $X_1, X_2, \dots, X_k$. This choice is valid, since $A \rightarrow X_1X_2 \cdots X_k$ is a production of G .

The node for each X_i is made the root of a subtree with yield w_i . In case (1), where X_i is a terminal, this subtree is a trivial tree with a single node labeled X_i . That is, the subtree consists of only this child of the root. Since $w_i = X_i$ in case (1), we meet the condition that the yield of the subtree is w_i .

In case (2), X_i is a variable. Then, we invoke the inductive hypothesis to claim that there is some tree with root X_i and yield w_i . This tree is attached to the node for X_i in Fig. 5.9.

The tree so constructed has root A . Its yield is the yields of the subtrees, concatenated from left to right. That string is $w_1w_2 \cdots w_k$, which is w . $\square$

From Trees to Derivation

Theorem 5.14: Let $G = (V, T, P, S)$ be a CFG, and suppose there is a parse tree with root labeled by variable A and with yield w , where w is in T^* . Then there is a leftmost derivation $A \xRightarrow[lm]{*} w$ in grammar G .

PROOF: We perform an induction on the height of the tree.

BASIS: The basis is height 1, the least that a parse tree with a yield of terminals can be. In this case, the tree must look like Fig. 5.8, with a root labeled A and children that read w , left-to-right. Since this tree is a parse tree, $A \rightarrow w$ must be a production. Thus, $A \xRightarrow[lm]{*} w$ is a one-step, leftmost derivation of w from A .

INDUCTION: If the height of the tree is n , where $n > 1$, it must look like Fig 5.9. That is, there is a root labeled A , with children labeled $X_1, X_2, \dots, X_k$ from the left. The X 's may be either terminals or variables.

1. If X_i is a terminal, define w_i to be the string consisting of X_i alone.
2. If X_i is a variable, then it must be the root of some subtree with a yield of terminals, which we shall call w_i . Note that in this case, the subtree is of height less than n , so the inductive hypothesis applies to it. That is, there is a leftmost derivation $X_i \xRightarrow[lm]{*} w_i$.

Note that $w = w_1 w_2 \cdots w_k$.

We construct a leftmost derivation of w as follows. We begin with the step $A \xRightarrow[lm]{*} X_1 X_2 \cdots X_k$. Then, for each $i = 1, 2, \dots, k$, in order, we show that

$$A \xRightarrow[lm]{*} w_1 w_2 \cdots w_i X_{i+1} X_{i+2} \cdots X_k$$

This proof is actually another induction, this time on i . For the basis, $i = 0$, we already know that $A \xRightarrow[lm]{*} X_1 X_2 \cdots X_k$. For the induction, assume that

$$A \xRightarrow[lm]{*} w_1 w_2 \cdots w_{i-1} X_i X_{i+1} \cdots X_k$$

- a) If X_i is a terminal, do nothing. However, we shall subsequently think of X_i as the terminal string w_i . Thus, we already have

$$A \xRightarrow[lm]{*} w_1 w_2 \cdots w_i X_{i+1} X_{i+2} \cdots X_k$$

From Trees to Derivation

- b) If X_i is a variable, continue with a derivation of w_i from X_i , in the context of the derivation being constructed. That is, if this derivation is

$$X_i \Rightarrow_{lm} \alpha_1 \Rightarrow_{lm} \alpha_2 \cdots \Rightarrow_{lm} w_i$$

we proceed with

$$\begin{aligned} w_1 w_2 \cdots w_{i-1} X_i X_{i+1} \cdots X_k &\Rightarrow_{lm} \\ w_1 w_2 \cdots w_{i-1} \alpha_1 X_{i+1} \cdots X_k &\Rightarrow_{lm} \\ w_1 w_2 \cdots w_{i-1} \alpha_2 X_{i+1} \cdots X_k &\Rightarrow_{lm} \\ \cdots & \\ w_1 w_2 \cdots w_i X_{i+1} X_{i+2} \cdots X_k & \end{aligned}$$

The result is a derivation $A \xRightarrow_{lm}^* w_1 w_2 \cdots w_i X_{i+1} \cdots X_k$.

When $i = k$, the result is a leftmost derivation of w from A . $\square$

Example

Try yourself.

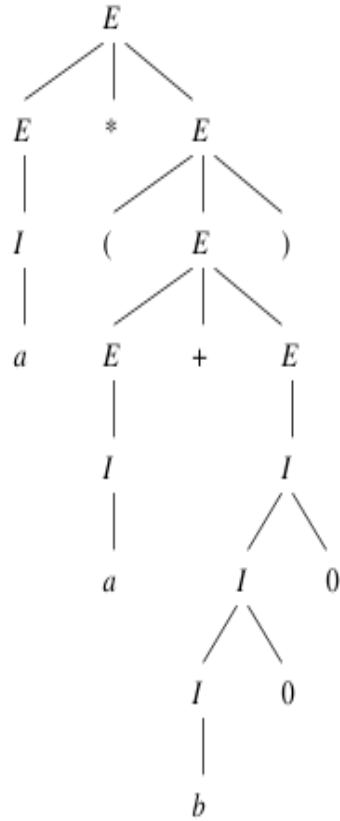


Figure 5.6: Parse tree showing $a * (a + b00)$ is in the language of our expression

- Section 5.2 of Chapter 5